TechPad: Start Here

A 12-key developer macropad for the Hack Club Hackpad challenge, by MoonRock. Project pack refreshed 19 Oct 2026.



The final design: key-well case, 98 × 93 × 22.4 mm.

Final spec

Part	Choice
Keys	12 MX switches, 4 across × 3 down, 19.05 mm pitch, white DSA keycaps
Knob	1 × EC11 encoder with push-click (the second knob was dropped to save pins)
Display	0.91" I²C OLED, pins GND-VCC-SCL-SDA, powered from 3V3
Lighting	12 × SK6812 MINI-E, one under each key, chained in a snake
Controller	Seeed XIAO RP2040, mounted on the back, USB-C at the top edge
Matrix	Duplex matrix: 3 + 3 pins drive 12 keys + knob click, 13 × 1N4148 diodes, no I/O expander
PCB	92 × 87 mm, 2 layers, GND fill on both sides
Case	Charcoal 3-piece print (tray, switch plate, top frame) with a key well that hides the switch bodies, 98 × 93 × 22.4 mm, no screws on top (inserts from below)

Where the project stands

Stage	Status
Plan and pin budget	✓ Done
Schematic	✓ Done, 0 ERC errors
PCB placement and routing	✓ Done, 0 DRC errors (18 Oct)
Silkscreen and ground fill	✓ Done

Gerbers and drill files	✓ Exported (zip them as gerbers.zip when you organise the folders)
Case (FreeCAD)	✱ **Now:** 01_Case_Guide_FreeCAD
Firmware (QMK, duplex matrix)	○ Next: Claude writes it, compiled to firmware.uf2
README and folder layout	○ Then: 04_Hackpad_Submission_Checklist
Hackpad submission	○ Then Hack Club reviews it and ships the parts
Print, solder, flash, fix fits	○ After the parts arrive: 05_After_Parts_Arrive
Desktop app (PySide6)	○ Last. Not needed for the submission

What is in this pack

File	Use it for
00_START_HERE.pdf	This overview
01_Case_Guide_FreeCAD.pdf	**Main guide.** Builds the whole case in FreeCAD, with every dimension
02_PCB_Reference.pdf	Pin map, part positions, design rules, how to export Gerbers and STEP
03_BOM_and_Hardware.pdf + BOM.csv	Every part, how many, and where it goes
04_Hackpad_Submission_Checklist.pdf	Exactly what to hand in, and the folder layout
05_After_Parts_Arrive.pdf	For later: measuring, fit test, fixes, printing, assembly, flashing
drawings/*.dxf (+ .png previews)	Case outlines at true position and height, ready to import into FreeCAD
images/	Design sheet, render, routed PCB, matrix diagram, schematic
reference/	Your schematic guide and the KiCad electronics guide (still valid)

I dropped the older notes pcb-placement.md, pcb-placement-kicad10.md, pcb-routing-guide.md and schematic-review-1.md. The finished board replaces them, and their positions now live in 02_PCB_Reference. The Monolith images were an earlier design direction, so they're left out too.

How this works: design first, build later

Hack Club sends the parts **after** you submit. So the whole design has to be finished first, with nothing measured or printed. That's normal for Hackpad. The case uses typical part sizes, and any small fits get fixed once the kit arrives.

Your next steps

Before you submit (design)

1. **Case in FreeCAD**: 01_Case_Guide_FreeCAD, sections 5 to 10. Skip anything about measuring or test printing.
2. **Export the STEP files**: Top, Bottom and Knob, plus the full assembly with pcb.step (guide section 10).
3. **Firmware**: QMK for the duplex matrix, knob, OLED and LEDs. Claude writes it and builds firmware.uf2. It can't be tested until the parts arrive.

4. **Screenshots**: the schematic, the PCB, the case, and the whole pad (the 3D preview link below works).
5. **Organise the folders and write the README**, then submit (04_Hackpad_Submission_Checklist).

After the parts arrive (build)

1. Measure the 8 parts and fix any fits (05_After_Parts_Arrive, sections 1 and 2).
2. Print the fit-test corner, then the full case.
3. Solder, flash, test, assemble.

3D preview of the agreed look: https://claude.ai/artifact/4LtL77XX4mXD5pFEnwg2R4